CLASS 9 • SST • EXAM WORKBOOK

GEOGRAPHY

the Workbook

Poora exam prep — samajhne wali **Hinglish notes**, likhne wale **English points**, saare **definitions**, **system trees** (iske under kya, uske under kya) aur full **practice set** (MCQs + big Q&A). All from your textbook, nothing missed.

2 **Shaping of the Earth's Surface**
Plate tectonics, weathering & erosion, agents of gradation — rivers, waves, glaciers, wind, underground water & landform disasters

3 **Atmosphere and Climate**
Composition & layers of atmosphere, weather vs climate, monsoon mechanism, seasons, climate change & the Punjab floods case study

💡 Hinglish Samajh · ✍️ Exam-Ready Points · 📖 Definition Bank · 🌳 System Trees · ⚖️ Differences · ✅ 10 + 10 MCQs · 📝 10 + 10 Big Q&A · 🎯 Under 20 Pages

Let's go! 🚀

1 · SAMAJH
Pehle 💡 Hinglish box padh — topic ka feel aa jayega

2 · RATTAAA
📖 Definition + ✍️ points yaad kar — yehi likhne ke hain

3 · TEST
MCQs khud solve kar, phir answers check kar 💪

4 · WRITE
Big Q&A dekh — points ko apne answers me copy-paryog karo

CHAPTER 2 • GEOGRAPHY

Shaping of the Earth's Surface



Earth ki surface kabhi fixed nahi hai — andar se powerful forces (earthquakes, volcanoes, folding, faulting) naye landforms banati hain, aur bahar se weathering + erosion unhe ghiste hain. Is chapter me: plate tectonics se lekar saare landforms aur disasters tak.

🔑 11 Topics | 📖 25+ Definitions | 🌳 5 System Trees | ✅ 10 MCQs | 📝 10 Q&A

📚 Notes — Har Topic, Poori Samajh

T1 Inside the Earth — 3 Layers 🌍

☐ revised

📖 DEFINITIONS

Crust — the outermost layer of the Earth on which we live. | **Mantle** — the very thick, hot, mostly solid layer between the crust and the core. | **Core** — the innermost, extremely hot and heavy layer of the Earth.

💡 SAMAJH (HINGLISH)

Socho ek kadu (pumpkin) ka structure — sabse bahar patli skin (crust), phir mota gudda (mantle), aur beech me sabse dense wala part (core). Hum crust pe rehte hain! Crust + upper mantle milke banta hai **lithosphere** (solid, rigid tile jaisa), aur uske neeche **asthenosphere** — jo semi-molten (aadha pighla hua) hai. Isi slippery asthenosphere pe plates slip-slide karti rehti hain!

🌳 SYSTEM — EARTH'S INTERIOR (ISKE UNDER YEH)

- **Crust (outermost)**
  - **Continental crust** — 30–40 km thick (under continents)
  - **Oceanic crust** — only 5–7 km thick (under oceans)
- **Mantle (~2900 km) → hot, mostly solid**
  - **Asthenosphere** — upper part, semi-molten (partially molten rock), lets plates move
- **Core (innermost, densest)**
  - **Outer core** (~2200 km) — fluid layer of iron and nickel
  - **Inner core** (~1250 km) — solid, hot, spinning metal ball
- **Lithosphere (~100 km) — crust + upper mantle; broken into tectonic plates**

✍️ EXAM POINTS

- The Earth is made of **three main layers: crust, mantle, and core** (Earth's radius ≈ 6375 km).
- **Lithosphere** = crust + upper mantle — the rigid outer layer; it is broken into **tectonic plates**.
- **Asthenosphere** = hot, mobile, semi-molten layer beneath the lithosphere — allows the plates to move.
- Crust thickness **varies**: 30–40 km under continents vs 5–7 km under oceans.

T2 Plate Tectonics — Theory 🧩

☐ revised

📖 DEFINITIONS

Plate tectonics — the theory (given by **W.J. Morgan**) explaining the movement of the Earth's crust: the outermost layer is broken into large and small pieces called **tectonic plates** that move slowly over the semi-molten layer beneath them. | **Landform** — a natural feature on the Earth's surface formed by weathering, erosion, deposition, or movement of the crust (mountains, valleys, plateaus, plains).

💡 **SAMAJH (HINGLISH)**

Socho Earth ka upper cover ek **tooti hui thali** jaisa hai — kai bade-bade tukde (plates) jo dheere-dheere (saal me bas **few centimetres**!) asthenosphere pe travel karte hain. Itna slow ki humein feel nahi hota, but crores of years me mountains khadi kar dete hain. Kaun dhakka deta hai? **Convection currents** — jaise garam paani upar uthta hai aur thanda neeche girta hai, waise hi mantle ka molten material!

🌳 **SYSTEM — TYPES OF TECTONIC PLATES**

- **Continental plates — carry continents**
- **Oceanic plates — carry ocean floors**
- **Mixed plates — carry both continents and oceans**

✍️ **EXAM POINTS — YEHI LIKHNA HAI**

- Theory given by **W.J. Morgan**; plates move **a few centimetres per year**.
- **Major plates:** Pacific, Eurasian, African, North American, South American, Indo-Australian, and Antarctic.
- Movement is caused by **convection currents in the mantle**: core's heat makes molten material **rise**, cooler material **sinks** — this continuous cycle pushes and pulls the plates.
- Plate movement gives rise to **mountains, valleys, ocean basins, volcanoes, earthquakes** and explains the distribution of continents and oceans.
- Helps identify **earthquake- and volcano-prone regions** and manage disasters.

T3 Plate Boundaries — 3 Types + Ring of Fire 🔥

☐ revised

📖 **DEFINITION**

Plate boundary — the edge where two tectonic plates meet. | **Ring of Fire** — the zone around the Pacific Ocean where most earthquakes and volcanoes occur (along plate boundaries).

💡 **SAMAJH (HINGLISH)**

Do plates ki "mulaqaat" ke 3 tareeke: **aana-jaana** (convergent — takkar!), **alag hona** (divergent — gap banta hai, magma bharta hai) aur **paas se guzarna** (transform — ragad se earthquakes). Pacific Ocean ke around plates ka junction hi hai **Ring of Fire** — isliye wahan sabse zyada earthquakes + volcanoes. India bhi risk me hai — 2001 me Gujarat ka bada bhukamp yaad rakho!

🌳 **SYSTEM — 3 PLATE BOUNDARIES (ISKE UNDER KYA HOTA HAI)**

- **Convergent — plates move towards each other**
  - Continental + continental → **fold mountains** (e.g., the Himalaya)
  - Oceanic + continental → oceanic plate **sinks beneath** (subduction) → **volcanoes + earthquakes**
- **Divergent — plates move away from each other**
  - Magma rises from below → forms **new crust** → creates **mid-ocean ridges** (e.g., the **Mid-Atlantic Ridge**)
- **Transform — plates slide past each other**
  - **No crust created or destroyed** → mainly causes **earthquakes** (e.g., the **San Andreas Fault**, USA)

✍️ **EXAM POINTS**

- Most earthquakes and volcanoes occur **along plate boundaries**, especially around the Pacific — the **Ring of Fire**.
- India has experienced major earthquakes (e.g., **Gujarat, 2001**) — a large quake in a densely populated country causes severe damage to life and environment.
- **Don't Miss Out:** Ancient India knew earthquakes as **bhūkampa**. In the **Bṛihatsaṁhitā**, **Varāhamihira** linked them to wind, rain, clouds and animal behaviour, attributing them to four forces — **Vāyu (wind), Agni (fire), Indra (thunder), Varuṇa (water)**.
- **Baratang Island** (Andaman & Nicobar) has India's only **mud volcano** — mud bubbles out due to underground gases and pressure.

T4 Weathering vs Erosion — Breaking vs Carrying 🪨

☐ revised

📖 **DEFINITIONS**

Weathering — the process through which rocks on the Earth's surface break down into smaller pieces; it does **not** involve movement of the broken material. | **Erosion** — the process by which soil, rocks and surface materials are worn away and **carried from one place to another** by natural agents like water, wind, ice or waves.

💡 **SAMAJH (HINGLISH)**

Ek line me: **Weathering = todna (bina hilaye)**, **Erosion = todna + utha ke le jaana**. Jaise exam me — question samajhna ≠ answer likh ke copy karna! Erosion me "transport" ka step extra hai. Dono milkar mountains ko ghisa kar plains banate hain — crore saal ka slow makeover 💅.

🌳 **SYSTEM — TYPES (DONO KE)**

- **Weathering — 3 types**
  - **Physical** — rocks break due to temperature changes, frost, or wind
  - **Chemical** — minerals change by reactions with water, air, or acids → new substances

- ▸ **Biological** — caused by plants, animals, micro-organisms (e.g., roots grow into cracks and split rocks)

- **Erosion — 4 types (by agent)**
  - ▸ **Water** (rivers, rain, ocean waves) · **Wind** (dry, sandy areas)
  - ▸ **Glacial** (moving ice scrapes and carries rocks) · **Coastal** (sea waves wear away the shore)

✍️ EXAM POINTS — EROSION KA ASAR + ANCIENT SOLUTIONS

- Weathering also helps in **forming soil**; erosion **shapes landforms** — can both create and destroy features.
- **Effect on occupations:** farmers lose fertile **topsoil** (lower yields); river/coast-side people lose land, houses, roads; construction and mining face unstable land; tourism and fishing suffer.
- **Ancient water & soil conservation (Sindhu-Sarasvatī civilisation):** contouring, bunding, terracing, dams and canals — recorded in the **Vedas, Kṛiṣhiparāśhara, Kauṭilya's Arthaśhāstra, Vṛikṣhāyurveda**.
- **Zabo system** (Nagaland) — earthen bunds on hillslopes for soil and water conservation; **check dams** across small streams slow water, stop erosion, recharge groundwater.
- Key terms: **Contour trench (CCT)** — trenches dug along contour lines of a hillside to hold rainwater; **Bunding** — earthen embankments along contour lines; **Terracing** — level steps cut on a hillside.

T5 Agents of Gradation — 5 Forces 🛠️

☐ revised

📖 DEFINITION

Agents of gradation — natural forces that wear down, transport, and deposit materials on the Earth's surface, helping to level or smooth it over time.

💡 SAMAJH (HINGLISH)

Gradation ka simple matlab — **"level karna"**. Earth ke upar-neeche (uneven) hisson ko dheere-dheere barabar karna. 5 kaam karne wale hain: **paani, baraf (glacier), hawa, lehrein (waves), aur zameen ke neeche ka paani (groundwater)** — upar wale areas ko neeche karte hain, neeche wale ko bhar dete hain. Erosion + deposition dono inke tool hain!

🌳 SYSTEM — 5 AGENTS & UNKA KAAM

- **Running water — erodes rocks and soils → forms valleys and plains**
- **Glaciers — scrape and carry huge material → carve U-shaped valleys**
- **Wind — shapes deserts by eroding and depositing sand**
- **Sea waves — erode coastlines → form cliffs, beaches, and bays**
- **Groundwater — dissolves rocks like limestone → creates caves and sinkholes**

✍️ EXAM POINTS

- Together these agents **continuously modify landforms** — lowering high areas and filling up low areas.
- Each agent makes its own **signature landforms** — ye next topics me ek-ek karke.

T6 Running Water — River's 3 Courses 🏞️

☐ revised

📖 DEFINITIONS

Waterfall — a landform where a river flows over a steep cliff or vertical drop; forms where **hard rocks resist erosion** while softer rocks below are worn away. | **Meander** — a winding curve or bend in the middle/lower course of a river, formed by **lateral erosion and deposition of sediments**. | **Delta** — a fan-shaped/triangular landform at the **mouth** of a river where it deposits sediments carried from upstream into a sea, ocean or lake.

💡 SAMAJH (HINGLISH)

Nadi ki life story 3 acts me: **Upper course** (jawaani) — full speed, ghaati kaat-ta hai (V-shaped valleys, waterfalls); **Middle course** (adulthood) — thak ke mudne lagti hai (meanders — jaise zameen pe 'S' likhta); **Lower course** (budhapa) — slow-slow, saara maal (sediment) samundar ke muh pe dump kar deti hai = **delta**! Sundarbans — India ka famous delta, super fertile.

🌳 SYSTEM — RIVER COURSE KE 3 STAGES (ISKE UNDER LANDFORMS)

- **Upper course — steep gradient, strong erosion**
  - ▸ **V-shaped valleys** · **waterfalls** (hard rock caps soft rock → sudden drop + plunge pool) · **rapids**
- **Middle course — loses energy, starts depositing**
  - ▸ **Meanders** (outer bank erodes, inner bank deposits) · **oxbow lakes** · **floodplains**
- **Lower course — slowest, maximum deposition**
  - ▸ **Deltas** · **levees** · **alluvial fans**

✍️ EXAM POINTS — HUMAN IMPORTANCE

- **Waterfall:** tourist attraction; **hydroelectric power** from the force of falling water; recreation (trekking, photography); cultural/religious significance in some regions.
- **Meander:** fertile soil along banks supports **agriculture**; villages/towns develop on gentle slopes near meanders; used for **navigation and irrigation** — e.g., the **Grand Anicut (Kallanai)** in Tamil Nadu, built for irrigation.

- **Delta:** rich **alluvial soil** → ideal for crops like **rice and jute**; fishing (fresh + saltwater mix); dense settlements, trade and transport routes — but **prone to flooding**.

📍 **India example: Sundarbans delta** — unique, popular with tourists; formed where the river meets the sea (distributaries, islands/bars).

T7 Waves & Currents — Coastal Work 🌊

☐ revised

📖 **DEFINITION**

Beach — a landform of sand, pebbles or rocks along the shoreline, created by the **deposition of sediments by waves**.

💡 **SAMAJH (HINGLISH)**

Lehrein coastal kaarigar hain — kuch **banati** hain (beach — sand jama karke), kuch **todti** hain (cliffs, caves). Order yaad rakho, exam me sequence poocha jata hai: waves pehle kamzor jagah **cave** khodti hain → do caves milke **arch** banati hain → arch girta hai to pillar bachta hai = **stack**! Jaise mitti ka ghar — pehle darwaza, phir mehraab, phir khamba.

🌳 **SYSTEM — COASTAL EROSION LANDFORMS (STEP-BY-STEP)**

- **Cliffs — steep rock faces as waves undercut the base of the coast**
- **Wave-cut platforms — flat areas left behind as cliffs retreat**
- **Caves — formed when waves erode weak parts of the rock**
- **Arches — created when caves on opposite sides of a headland meet**
- **Stacks — isolated pillars of rock left standing after arches collapse**

✍️ **EXAM POINTS**

- Beaches are shaped constantly by **wave action, tides and currents**.
- **Beach importance:** tourism (boosts local economy); fishing areas; sand and shell collection; act as **natural barriers** protecting coastal settlements from strong waves and erosion.
- Some coastal areas need **coastal protection** to safeguard settlements; others are tourism hubs.

T8 Glaciers — Baraf ka Bulldozer 🧊

☐ revised

📖 **DEFINITION**

Moraines — landforms created by the **deposition of till** (rocks, soil, debris) carried along and left behind by glaciers when they melt.

💡 **SAMAJH (HINGLISH)**

Glacier = **slow-moving ice ki nadi**. Jaise bulldozer road pe mitti uthata-dhakelta hai, glacier pahadon ko scrape karke **U-shaped valley** banata hai (nadi V-shape banati hai, glacier U — yaad rakhna!). Pighalne pe saara jama kiya maal (till) chhod deta hai = **moraines** — side me (lateral), end pe (terminal), aur do glaciers milne pe beech me (medial).

🌳 **SYSTEM — GLACIAL LANDFORMS**

- **Erosion wale**
  - **U-shaped valleys** — glaciers widen and deepen river valleys
  - **Cirques** — bowl-shaped depressions at the head of a glacier
  - **Aretes** — sharp ridges between valleys
  - **Hanging valleys** — where smaller glaciers meet larger ones
  - **Fjords** — deep, narrow inlets when the sea floods glacial valleys
- **Deposition wale (moraines)**
  - **Lateral** — along the sides of glaciers
  - **Terminal** — at the end, marking the glacier's furthest advance
  - **Medial** — in the middle, when two glaciers meet and lateral moraines join

✍️ **EXAM POINTS — HUMAN IMPORTANCE**

- U-shaped valleys and cirques → **trekking, skiing, mountaineering** tourism.
- **Fjords** are used as harbours and for fishing; some valleys have fertile glacial soil for agriculture.
- Glaciers are crucial **sources of fresh water** — feeding rivers that sustain populations downstream.
- Moraines often create **fertile soil**, natural dams and lakes — used for water supply, irrigation, hydroelectric power.

T9 Wind + Underground Water 🏜️

☐ revised

📖 **DEFINITIONS**

Dunes — hills or ridges of sand formed by wind in deserts or along sandy coasts. | **Karst topography** — unique landforms created by underground water through **chemical weathering and erosion of limestone** (soluble rocks). | **Stalactites** — icicle-shaped formations hanging from cave **ceilings**; **Stalagmites** — formations rising from cave **floors** (when both join → a pillar).

💡 **SAMAJH (HINGLISH)**

Desert me hawa boss hai — sand ko udha kar jahan girti hai wahan **dunes**; jahan khodti hai wahan **deflation hollows** (aur paani ki table tak pahunch jaye to **oasis** — desert ka swimming pool 🏝️). Zameen ke neeche groundwater **limestone ko gholta** hai (jaise namak paani me) — ban jaate hain **caves**, aur andar mineral-laden drops se **stalactite** (ceiling se latakti — "c" for ceiling!) aur **stalagmite** (floor se khadi — "g" for ground!).

🌳 **SYSTEM — WIND LANDFORMS + KARST (UNDERGROUND WATER) LANDFORMS**

- **Wind — erosion landforms**
  - **Yardangs** — streamlined rock ridges carved by wind
  - **Ventifacts** — rocks polished and shaped by sandblasting
  - **Deflation hollows / blowouts** — shallow depressions where loose material is removed (may expose water table → **oasis**)
  - **Desert pavements** — flat surfaces left after finer particles are blown away
- **Wind — deposition: 4 dune types**
  - **Barchan** — crescent-shaped; limited sand + single wind direction
  - **Longitudinal** — long ridges parallel to the prevailing wind
  - **Star** — multiple arms; winds from different directions
  - **Parabolic** — U-shaped, often stabilised by vegetation
- **Karst (underground water)**
  - **Caves** — hollow spaces as acidic water dissolves rock
  - **Stalactites** (ceiling) & **stalagmites** (floor) · **pillars** (when they join)
  - **Sinkholes / dolines** — depressions when ground collapses into an underground cavity
  - **Underground rivers** — flow through cave systems

✍️ **EXAM POINTS**

- Wind landforms influence **settlement and agriculture** in arid regions; yardangs and ventifacts attract **tourists and geologists**.
- **Dunes:** natural barriers against desertification and wind erosion; protect coastal settlements from sea winds; sand sometimes used for construction.
- **Caves & underground rivers:** fresh water sources, tourism, cultural/religious significance; stalagmites–stalactites attract geologists and adventurers.

T10 Landform Disasters — 4 Villains ⚠️

☐ revised

📖 **DEFINITIONS**

Landslide — sudden slope failure when soil and rocks slide down unstable slopes. | **Avalanche** — sudden collapse of snow down steep mountain slopes. | **GLOF (Glacial Lake Outburst Flood)** — sudden release of large volumes of water from glacial lakes when their natural dams (ice or loose moraines) collapse. | **Dust storm** — strong winds lifting large amounts of loose, dry soil and sand into the air.

💡 **SAMAJH (HINGLISH)**

Har landform ke saath uska disaster bhi attached hai: **pahad** → landslide/avalanche; **glacier** → GLOF (glacier pighla, lake bani, moraine dam toota — poora paani ek jhatke me neeche!); **desert** → dust storm. Causes hamesha 2 type ki hain: **natural** (baarish, earthquake, heat) + **human-made** (deforestation, mining, unplanned construction). Yehi trick yaad rakho, answer me dono likhoge!

🌳 **SYSTEM — 4 DISASTERS & THEIR CAUSES (NATURAL + HUMAN)**

- **Landslide**
  - Natural: heavy continuous rainfall (water seeps in → weight ↑, friction ↓); earthquakes and volcanic eruptions; steep slopes; loose/weathered rocks
  - Human: deforestation, mining, road construction, unplanned hillside construction; poor drainage; improper land use
- **Avalanche**
  - Natural: heavy snowfall in a short time (extra weight); sudden temperature rise (partial melting → less friction); strong winds piling snow unevenly; earthquakes and vibrations
  - Human: skiing, trekking, construction in mountainous areas disturbing snow-covered slopes
- **GLOF**
  - Rapid glacier melting (rising temperatures) → lake size and pressure ↑ on ice/moraine dams; heavy rain or snowfall adds water; earthquakes, avalanches or landslides weaken the dam → sudden collapse → destructive downstream floods
- **Dust storm**
  - Natural: prolonged drought and low rainfall → loose, dry soil; common in desert and semi-arid regions

- Human: sparse vegetation from deforestation, overgrazing, poor farming; climate change increases frequency and intensity

✍️ **EXAM POINTS**

- **Chamoli, Uttarakhand (Feb 2021):** devastating flood — people and livestock lost lives; severe damage to buildings, roads, bridges and hydel projects; village connectivity affected.
- For each disaster, be ready to list: **prone areas + mitigation measures + recent examples** (book asks you to research these).

T11 Landforms & Human Life + Final Recap 🌍

☐ revised

💡 **SAMAJH (HINGLISH)**

History + geography ka connection: **rewa-sabun jaisi fertile plains** (Ganga, Nile, Brahmaputra, Indus) ne agriculture aur early cities di; **Himalaya** ne India ko invasions se bachaya but Khyber pass se cultural exchange bhi; **Thar desert** ne badi settlements nahi di, but Silk Route jaise trade routes; **coasts** ne trade, travel aur cultural contacts diye — south India ke kingdoms isliye flourish hue. Simple: jaisi zameen, waisi kahani!

✍️ **EXAM POINTS — BEFORE WE MOVE ON (BOOK SUMMARY)**

- The Earth is made of layers — **crust, mantle, core**.
- **Internal forces** (earthquakes, volcanoes, folding, faulting) move the crust and build landforms; **external forces** (weathering, erosion, deposition) carve smaller landforms.
- The Earth's surface is carved by agents of gradation — **running water, waves and tides, glaciers, wind, underground water**.
- Disasters like **landslides, avalanches, GLOFs, dust storms** are associated with specific landforms and affect human life in multiple ways.
- Human life is deeply connected to landforms — they influence our **climate, resources, settlements, and cultures**.

⚖️ Differences — Pakka Aane Wale

Basis	Weathering	Erosion
Meaning	Breakdown of rocks into smaller pieces on the Earth's surface	Wearing away of soil, rocks and surface materials
Movement	**No movement** of broken material — only breaking down	Broken material is **carried from one place to another**
Types	Physical, chemical, biological	Water, wind, glacial, coastal erosion
Role	Breaks rocks; helps form **soil**	Shapes landforms; can create and destroy features

Basis	Internal Forces	External Forces
Source	Inside the Earth (heat of the interior)	On/at the surface (sun-driven agents)
Examples	Earthquakes, volcanic eruptions, folding, faulting	Weathering, erosion, deposition
Effect	Build mountains, valleys, ocean basins (make the surface uneven)	Wear down and reshape landforms (level the surface)
Speed	Sudden and drastic (minutes to years)	Slow and gradual (thousands of years)

Basis	Stalactite	Stalagmite
Position	Hangs from the **ceiling** of a cave	Rises from the **floor** of a cave
Shape	Icicle-shaped (pointing down)	Grows upward like a column
Both join	Together they form a **pillar** inside the cave (Karst topography)	

📖 Definition Bank — Chapter 2 (likhne ke liye ready)

Landform — natural feature on the Earth's surface formed by weathering, erosion, deposition or crustal movement.

Tectonic plates — massive slabs of solid rock moving slowly (a few cm/year) over the semi-molten layer.

Ring of Fire — zone around the Pacific Ocean with most earthquakes and volcanoes.

Asthenosphere — hot, mobile, semi-molten layer beneath the lithosphere.

Erosion — wearing away and carrying away of surface material by water, wind, ice or waves.

Waterfall — river flowing over a steep cliff; hard rock resists, soft rock below is worn away.

Delta — fan-shaped deposition at the mouth of a river (e.g., Sundarbans).

Moraines — depositional landforms of till (glacier debris): lateral, terminal, medial.

Dunes — sand hills/ridges made by wind: barchan, longitudinal, star, parabolic.

Stalactite / Stalagmite — cave formations from ceiling / from floor.

Avalanche — sudden slide of unstable snow on steep slopes.

Dust storm — strong winds lifting loose, dry soil and sand into the air.

Bunding — earthen embankments along contour lines to slow run-off.

Plate tectonics — theory by W.J. Morgan explaining the movement of the Earth's crust broken into tectonic plates.

Plate boundary — the edge where two tectonic plates meet (convergent, divergent, transform).

Lithosphere — rigid outer layer = crust + upper mantle; broken into tectonic plates.

Weathering — breakdown of rocks into smaller pieces without movement.

Agents of gradation — running water, glaciers, wind, waves, groundwater.

Meander — winding bend formed by lateral erosion and deposition.

Beach — coastal deposition of sand, pebbles or rocks by waves.

Till — rocks, soil and debris carried along and deposited by glaciers.

Karst topography — landforms made by underground water dissolving limestone.

Landslide — sudden slope failure; natural + human causes.

GLOF — Glacial Lake Outburst Flood; sudden release of glacial lake water.

Contouring (CCT) — trenches along contour lines to hold rainwater and stop soil erosion.

Terracing — level steps cut on a hillside to prevent soil erosion.

✅ MCQ Challenge — 10 Questions (Ch 2)

🎯 Solve karne ke baad score bharo → ____ / 10

Q1. The theory of plate tectonics was given by —

a) Alfred Wegener
b) W.J. Morgan
c) Varāhamihira
d) Kauṭilya

Q2. The lithosphere consists of —

a) the crust only
b) the crust and the upper mantle
c) the mantle and the core
d) the whole mantle

Q3. The movement of tectonic plates is caused by —

a) rotation of the Earth
b) gravity of the Moon
c) convection currents in the mantle
d) ocean tides

Q4. When two continental plates collide, they form —

a) mid-ocean ridges
b) fold mountains like the Himalaya
c) transform faults
d) ocean trenches only

Q5. Most earthquakes and volcanoes occur around the Pacific Ocean in a zone called the —

a) Ring of Fire
b) Ring of Water
c) Circle of Rocks
d) Belt of Ice

Q6. Which of the following involves no movement of broken material?

a) Erosion
b) Deposition
c) Weathering
d) Transportation

Q7. Plant roots growing into rock cracks and splitting them apart is an example of —

a chemical weathering
b physical weathering
c biological weathering
d glacial erosion

Q8. U-shaped valleys, cirques, aretes and fjords are created by —

a rivers
b wind
c sea waves
d glaciers

Q9. The sequence of coastal erosion landforms when a headland is worn away is —

a stack → arch → cave → cliff
b cave → arch → stack
c arch → cave → beach
d beach → bar → stack

Q10. A Glacial Lake Outburst Flood (GLOF) happens when —

a a river delta silts up
b a dam is built on a river
c a glacial lake's ice/moraine dam suddenly collapses
d heavy snow falls on a plain

🔑 Answer Key — Chapter 2

1 → b | 2 → b | 3 → c | 4 → b | 5 → a | 6 → c | 7 → c | 8 → d | 9 → b | 10 → c

- **1:** Plate tectonics theory = W.J. Morgan (earth science).
- **2:** Lithosphere = crust + upper mantle (≈100 km thick, broken into plates).
- **3:** Core's heat → molten material rises, cooler sinks → convection currents push/pull plates.
- **4:** Continental–continental convergence (convergent boundary) → fold mountains (Himalaya).
- **5:** Ring of Fire = Pacific rim zone with most quakes + volcanoes (along plate boundaries).
- **6:** Weathering = breakdown WITHOUT movement; erosion involves movement.
- **7:** Plants/animals/micro-organisms = biological weathering.
- **8:** Glaciers widen valleys into U-shapes; cirques, aretes, hanging valleys, fjords all glacial.
- **9:** Waves erode weak rock → caves; caves on opposite sides meet → arch; arch collapses → stack.
- **10:** GLOF = sudden release of water when the glacial lake's natural (ice/moraine) dam breaks.

📝 Big Questions — Exam-Ready Answers (Ch 2)

Q1. Describe the layers of the Earth's interior.

ANSWER

- The Earth is made up of **three main layers: the crust, the mantle, and the core**.
- The **crust** is the outermost layer on which we live — 30–40 km thick under continents but only 5–7 km under the oceans.
- The **mantle** lies below the crust — very thick, hot and mostly solid; the upper semi-molten part is the **asthenosphere**.
- The **core** is the innermost and densest layer — the outer core is a fluid layer of iron and nickel; the inner core is a solid, hot, spinning metal ball.
- The crust together with the upper mantle forms the **lithosphere**, which is broken into tectonic plates that move over the asthenosphere.

Q2. What is plate tectonics? What causes the movement of tectonic plates?

ANSWER

- Plate tectonics is an important theory in earth science given by **W.J. Morgan** explaining the movement of the Earth's crust.
- According to it, the outermost layer is not one single piece but is broken into several large and small pieces called **tectonic plates** (continental, oceanic, or mixed).
- These plates move very slowly — usually **a few centimetres per year** — over the semi-molten layer beneath them.
- The movement is caused by **convection currents in the mantle**: heat from the Earth's core makes molten material rise while cooler material sinks, and this continuous cycle pushes and pulls the plates.
- Major plates include the Pacific, Eurasian, African, North American, South American, Indo-Australian and Antarctic plates.

Q3. Explain the three types of plate boundaries with their effects.

ANSWER

- **Convergent boundary** — plates move towards each other: continental plates colliding form **fold mountains** like the Himalaya; an oceanic plate colliding with a continental plate **sinks beneath it**, leading to volcanic activity and earthquakes.

- **Divergent boundary** — plates move away from each other: **magma rises from below** and forms new crust, creating features like **mid-ocean ridges** — e.g., the Mid-Atlantic Ridge.
- **Transform boundary** — plates **slide past each other** without creating or destroying crust, mainly causing **earthquakes** — e.g., the San Andreas Fault in the USA.
- Most earthquakes and volcanoes occur along these plate boundaries, especially around the Pacific Ocean, known as the **Ring of Fire**.

Q4. Distinguish between weathering and erosion. How does erosion affect human occupations?

ANSWER

- **Weathering** is the breakdown of rocks into smaller pieces **without any movement** of the material — physical, chemical, or biological.
- **Erosion** is the wearing away of soil and rocks that **involves movement**, by agents like water, wind, ice, or waves.
- **Effect on occupations:** for farmers, erosion removes the fertile **topsoil**, lowering crop yields.
- People near rivers and coasts lose **land, houses and roads**; in construction and mining, erosion destabilises land and poses safety risks.
- Industries like **tourism and fishing** suffer when beaches, rivers and fertile lands are destroyed.

Q5. What are agents of gradation? Name the landforms created by each.

ANSWER

- Agents of gradation are **natural forces that wear down, transport and deposit materials**, levelling the Earth's surface over time.
- **Running water** — erodes rocks and soils to form valleys and plains (waterfalls, meanders, deltas).
- **Glaciers** — scrape and carry material, carving **U-shaped valleys**, cirques, aretes, fjords, moraines.
- **Wind** — shapes deserts: yardangs, ventifacts, deflation hollows, desert pavements, **dunes**.
- **Sea waves** — erode coastlines into cliffs, caves, arches, stacks, and deposit **beaches**.
- **Groundwater** — dissolves limestone creating **Karst topography**: caves, stalactites, stalagmites, sinkholes, underground rivers.

Q6. Explain how a river creates different landforms in its three courses.

ANSWER

- Rivers shape the land through **erosion, transportation and deposition** along their course.
- **Upper course:** steep gradients and strong erosive forces create **V-shaped valleys, waterfalls and rapids** — waterfalls form where hard rocks resist erosion while softer rocks below are worn away.
- **Middle course:** the river begins to **meander** — eroding outer banks and depositing sediment on inner banks; forms **oxbow lakes and floodplains**.
- **Lower course:** the slow river deposits large sediment loads, forming **deltas, levees and alluvial fans** at its mouth.
- These landforms support agriculture (fertile banks), settlements, navigation and irrigation — e.g., the **Grand Anicut (Kallanai)** in Tamil Nadu.

Q7. How are coastal landforms created by waves and currents?

ANSWER

- Waves and currents constantly move over the ocean surface and **reshape the coastal zone**.
- **Deposition:** beaches are formed by sediments deposited by waves along the shoreline; they are popular tourist spots and act as natural barriers against strong waves.
- **Erosion:** waves undercut the coast to form **cliffs**; flat **wave-cut platforms** are left behind as cliffs retreat.
- Waves erode weak rock to form **caves**; caves on opposite sides of a headland meet to form **arches**; when arches collapse, isolated rock pillars called **stacks** remain.
- These landforms influence tourism and may require **coastal protection** for settlements.

Q8. Describe the landforms made by glaciers.

ANSWER

- **Erosional landforms: U-shaped valleys** (glaciers widen and deepen river valleys), **cirques** (bowl-shaped depressions at a glacier's head), **aretes** (sharp ridges between valleys), **hanging valleys** (where smaller glaciers meet larger ones) and **fjords** (deep, narrow inlets formed when the sea floods glacial valleys).
- **Depositional landforms — moraines:** formed of **till** (rocks, soil, debris) left behind by melting glaciers.
- Types: **lateral** (along the sides), **terminal** (at the end, marking the furthest advance) and **medial** (in the middle, when two glaciers' lateral moraines join).
- **Human importance:** tourism (trekking, skiing, mountaineering), fjords as harbours and fishing bases, fertile glacial soil, and moraine-formed dams and lakes for water supply, irrigation and hydroelectric power.
- Glaciers are also crucial **sources of fresh water** feeding rivers downstream.

Q9. Write a short note on the landforms created by underground water.

ANSWER

- Underground water, especially in **limestone or soluble rock** areas, creates **Karst topography** through chemical weathering and erosion.
- **Caves** — hollow spaces formed as acidic water dissolves rock.
- **Stalactites** — icicle-shaped formations hanging from cave ceilings; **stalagmites** — formations rising from the floors; when both join, a **pillar** is formed.
- **Sinkholes (dolines)** — depressions formed when the ground collapses into an underground cavity; **underground rivers** flow through cave systems.
- These provide fresh water, tourism opportunities, and cultural or religious significance.

Q10. What causes landslides? How do human activities increase their risk?

ANSWER

- **Natural causes:** heavy and continuous rainfall seeps into soil and rocks, increasing weight and reducing friction; earthquakes and volcanic eruptions shake and weaken slopes; steep slopes and loose, weathered rocks increase risk.
- **Human activities:** deforestation, mining, road construction and unplanned construction on hillsides disturb the natural balance of slopes.
- Poor drainage systems and improper land use allow excess water to accumulate, leading to **sudden slope failure**.
- Avalanches, GLOFs and dust storms are other landform-related disasters — e.g., the **Chamoli (Uttarakhand) flood of February 2021** damaged buildings, roads, bridges and hydel projects.
- Mitigation needs awareness, planning and protecting slopes (as done in ancient India through contouring, bunding and terracing).

CHAPTER 3 • GEOGRAPHY

Atmosphere and Climate



Asman hawa ka ek blanket hai — atmosphere. Isme kaun-kaun si gases hain, kitni layers hain, weather climate se alag kaise hai, monsoon ka full mechanism, seasons, climate change aur Punjab floods — sab is chapter me, zero se!

🔑 10 Topics | 📖 25+ Definitions | 🌳 4 System Trees | ✅ 10 MCQs | 📝 10 Q&A

📚 Notes — Har Topic, Poori Samajh

T1 Atmosphere — Hawa ka Blanket 🌫️

☐ revised

📖 DEFINITIONS

Atmosphere — a blanket of air surrounding the Earth, pulled down around it by gravity; a mixture of gases vital for the survival of all living beings. | **Gravity** — a fundamental physical force of attraction between objects that have mass or energy (e.g., the Sun and the Earth).

💡 SAMAJH (HINGLISH)

Earth ko socho ek **cocoon me lipda planet** — wahi atmosphere hai. Gravity isko Earth ke pass khichti rehti hai (warna hawa space me ud jaati). 3 kaam: (1) Sun ke **harmful radiation (UV) se shield**, (2) **temperature regulate** karti hai — Sun ki energy trap karke wapas space me jaane se rok-ti hai, (3) **weather aur climate** ka control room. Bina atmosphere ke — din pe garmi, raat pe barf, aur koi paani nahi!

🌳 SYSTEM — COMPOSITION OF THE ATMOSPHERE (%)

- **Nitrogen — 78% (most abundant)**
- **Oxygen — 21% (second most abundant)**
- **Argon — 0.93%**
- **Carbon dioxide — 0.04%**
- **Others — 0.03% — helium, neon, krypton, xenon, ozone, hydrogen**
- **Water vapour — 0.1% to 0.4% (varies) — plays a significant role in cloud formation and precipitation; plus tiny dust particles**

✍️ EXAM POINTS

- **Nitrogen and oxygen** are the two primary and most abundant gases essential for life.
- The composition of the atmosphere **varies with altitude**.
- For the pie diagram question: draw a circle — 78% N_2, 21% O_2, small slices for Ar 0.93%, CO_2 0.04%, others.

T2 Layers of the Atmosphere — 5 Floors 🏢

☐ revised

📖 **DEFINITIONS**

Altitude — the height of a location above mean sea level (sea level = zero). Layers are defined on the basis of changes in **temperature and density** with altitude — air density is highest near the surface and decreases with altitude.

💡 **SAMAJH (HINGLISH)**

Atmosphere = 5-manzila building, floor change hone par temperature ka behaviour badal jaata hai: **Troposphere** (ground floor — hum sab yahan; jitna upar, utni thand, yahi saara weather hota hai), **Stratosphere** (2nd — aeroplanes yahan udte hain kyunki clouds/td nahi; ozone ka guard room), **Mesosphere** (3rd — shooting stars yahan jalti hain ✨), **Thermosphere** (4th — garam! radio waves bounce + aurora ka jhilmil), **Exosphere** (roof — itna patla ki hawa space me nikal jaati hai).

🌳 **SYSTEM — 5 LAYERS, BOTTOM SE TOP (HEIGHT + KAAM)**

- **Troposphere — avg height ~12 km; temperature decreases with altitude**
  - The air we breathe + most water vapour and clouds; **nearly all weather** (rainfall, fog, hail)
  - Separated from stratosphere by the **tropopause**
- **Stratosphere — up to 50 km; ideal for flying aeroplanes (free of clouds and weather)**
  - Contains the **ozone layer** — filters the Sun's harmful ultraviolet radiation
  - Boundary above: **stratopause**
- **Mesosphere — up to 80 km; temperature decreases with altitude**
  - Most **meteorites burn up** here
- **Thermosphere — 80 to 700 km; temperature rises very rapidly (gas molecules absorb X-rays and short-wave UV)**
  - Helps **radio transmission** by reflecting radio waves back to Earth; the **ionosphere** is part of it
  - **Auroras** (northern and southern lights) occur here
- **Exosphere — uppermost; very thin air — light gases (helium, hydrogen) float into space due to weak gravity**

✍️ **EXAM POINTS**

- **Don't Miss Out:** temperature decreases with altitude **only in the troposphere and mesosphere**.
- **Aurora:** word from Latin, meaning 'dawn' (Roman goddess Aurora); **solar wind** (charged particles from the Sun) is directed to the magnetic poles, and each gas glows a particular colour — **Aurora Borealis** (Northern Hemisphere), **Aurora Australis** (Southern Hemisphere).
- **Don't Miss Out:** we don't feel air pressure because air presses from all sides and our bodies exert an **equal counter-pressure**.

T3 Weather vs Climate + 5 Elements 🌡️

☐ revised

📖 **DEFINITIONS**

Weather — the hour-to-hour and day-to-day conditions of the atmosphere. | **Climate** — the sum total of weather conditions and variations over a large area for an extended period of time, usually **thirty years or more**.

💡 **SAMAJH (HINGLISH)**

Aaj ka mood = **weather** (aaj baarish, kal dhoop); 30+ saal ka average nature = **climate** (Mumbai ka climate humid hai, Rajasthan ka dry). Simple — "weather aaj ka, climate zindagi bhar ka!" Weather banane ke 5 elements hain, jo humare chapter ke heroes hain: **temperature, precipitation, humidity, wind, atmospheric pressure**.

✍️ **EXAM POINTS — 5 ELEMENTS OF WEATHER & CLIMATE**

- **Temperature:** varies day–night and across seasons; key factor = **insolation** (incoming solar energy intercepted by the Earth) — insolation decreases from the equator towards the poles, so temperature also **decreases from equator to poles**.
- **Humidity:** presence of water vapour in the air; as air gets warmer, its capacity to hold water vapour increases — on humid days clothes take longer to dry and sweat evaporates slowly (discomfort!).
- **Precipitation:** when part of the atmosphere is saturated with water vapour which condenses and falls due to gravity — includes **drizzle, rain, snow, sleet, hail**; affected by prevailing winds, mountains, and seasons; rain is the most common form; long absence of rain → dry climate; most groundwater is collected from rainwater.
- **Atmospheric pressure:** pressure exerted by the weight of air — **highest at sea level, falls with altitude**; horizontally, it depends on temperature: warm air rises → **low pressure** (cloudy, wet weather); cold air sinks → **high pressure** (clear, sunny skies).
- **Wind:** movement of air from **high-pressure to low-pressure areas**.

T4 Wind — Speed Table + Naam ka Rule 💨

☐ revised

💡 **SAMAJH (HINGLISH)**

Hawa ka funda ek line me: **"pressure gap hua toh hawa bhaagi"** — high pressure se low pressure ki taraf. Naam rakhne ka rule twist hai: winds ko un **direction ke naam par** jaana jaata hai jahan **se** aati hain! West se aayi toh **westerly**. Confusing? Trick: "FROM wali direction = naam". Aur pressure-direction diagram — garam jagah (low pressure) pe clouds, thandi jagah (high pressure) pe clear sky.

Wind (category)	Speed (km/hr)	Common Effects
Calm	0–1	Calm; smoke rises vertically.
Light breeze	6–11	Wind felt on the face; leaves rustle; ordinary vane moves.
Strong breeze	39–49	Large branches sway; umbrellas difficult to use.
Storm	103–117	Rarely experienced; usually accompanied by widespread damage.

✍️ **EXAM POINTS**

- Wind can be gentle or strong — from blowing dry leaves to **uprooting trees** during a storm.
- **Don't Miss Out:** winds are named after the direction they **blow from** — e.g., the wind blowing from the west is called **westerly**.
- Air always moves from **high-pressure areas to low-pressure areas**; its direction is influenced by the Earth's rotation.

T5 Land Breeze vs Sea Breeze 🌅

☐ revised

📖 **DEFINITIONS**

Land breeze — a local wind that blows **from land to sea during the night**, due to differential cooling — land cools faster than the sea at night, so the temperature and pressure difference is low and wind speed is low. | **Sea breeze** — a local wind that blows **from sea to land during the day** (especially the afternoon), when the land becomes relatively warmer than the sea, forming a low-pressure region over land.

💡 **SAMAJH (HINGLISH)**

Beach trip wala logic: **Din me zameen tez garam** hoti hai (sand jalata hai!) → zameen pe low pressure → hawa **samundar se zameen** pe aati hai = sea breeze — isliye beach pe dopahar me mast thandi hawa! **Raat me zameen tez thandi** → zameen pe high pressure → hawa **zameen se samundar** = land breeze. Dono coastal areas ka **natural AC** hain — moderate climate banate hain.

✍️ **EXAM POINTS**

- Local winds like land and sea breezes affect the **weather and climate of coastal regions** and create **moderate climatic conditions** there.
- **Day:** land warmer than water → low pressure over land → wind blows sea → land.
- **Night:** land cools faster than the sea → pressure difference is small → gentle wind blows land → sea.

T6 Seasons in India — 4 (IMD) + 6 (Traditional) 🗓️

☐ revised

📖 **DEFINITION**

Seasons — divisions of the year based on generalised monthly atmospheric conditions or a common pattern over a few weeks or months (cool or hot, windy or calm, cloudy or bright, wet or dry).

💡 **SAMAJH (HINGLISH)**

IMD (Indian Meteorological Department) ke hisaab se India ka saal **4 seasons** me bata hai: **Winter** (Dec–early Apr), **Summer** (Apr–Jun), **Monsoon** (Jun–Sep), **Post-monsoon** (Oct–Dec). Lekin humare traditional calendar me **6 ṛtu** hain — Vasanta (bahaar), Grīṣhma (garmi), Varṣhā (baarish), Śharad (patjhad ke baad), Hemanta, Śhiśhira (sardi). Himalayan states (thandi wali) me autumn aur spring ke saath **6 seasons** dikhte hain!

🌳 **SYSTEM — IMD KE 4 SEASONS (MONTHS + DETAILS)**

- **Winter — December to early April**
  - Coldest months: **December–January**; average temperature ~10–15°C in the north-west, ~20–25°C in the south-east
- **Summer / pre-monsoon — April to June (up to July in north-west India)**
  - Hottest month: **April** in western & southern regions, **May** in northern regions; average 32–40°C across inland India
- **Monsoon / rainy (advancing monsoon) — June to September**
  - Humid **south-west summer monsoon** sweeps across the country from late May/early June; recedes from north India from early October; **South India receives more rainfall** during this retreat
- **Post-monsoon (retreating monsoon) — October to December**
  - North-west India: October–November usually **cloudless**

Season (Ṛtu)	Indian Calendar Months	Gregorian Months
Vasanta (spring)	Chaitra–Vaiśākha	March–April
Grīṣhma (summer)	Jyeṣṭha–Āṣhāḍha	May–June
Varṣhā (monsoon)	Śhrāvaṇa–Bhādrapada	July–August
Śharad (autumn)	Āśhvina–Kārtika	September–October
Hemanta (pre-winter)	Mārgaśhīrṣha–Pauṣha	November–December
Śhiśhira (winter)	Māgha–Phālguna	January–February

✍️ **EXAM POINTS**

- The climate of India is broadly classified as **tropical monsoon**.
- Traditional six seasons are based on the **astronomical division of 12 months into six parts**; each is about two months long (durations vary across regions).
- **Don't Miss Out:** Kauṭilya's **Arthaśhāstra** contains records of scientific measurements of rainfall and their use in managing revenue and relief efforts.
- Himalayan states, being more temperate, experience **two additional seasons — autumn and spring**.

T7 Monsoon — Full Mechanism ☔

☐ revised

📖 **DEFINITION**

Monsoon — the **seasonal reversal in wind direction** during a year. The word comes from the Arabic **mausim**, meaning 'season' — Arab traders named it after noticing the reversal of winds on their sailing voyages to India.

💡 **SAMAJH (HINGLISH)**

Monsoon = **winds ka direction ulta hona**, season ke saath! Story: **Garmi (June–Sept)** — India ki zameen samundar se tez garam → zameen pe **LOW pressure**, samundar pe high → **geeli hawa samundar se zameen** pe aati hai = **South-West Monsoon** → baarish! 💧 **Sardi (Oct–Feb)** — zameen tez thandi → zameen pe **HIGH pressure** → **thandi sookhi hawa zameen se samundar** = **North-East Monsoon**. Twist: ye hawa **Bay of Bengal cross** karti hai to moisture utha leti → **Tamil Nadu, Andhra Pradesh, Karnataka** ke east coast pe baarish! Isliye Chennai ki baarish saal ke end me hoti hai jab baaki India sookha ho chuka hota hai.

🌳 **SYSTEM — 2 MONSOONS (DIRECTION + MONTHS + KAAM)**

- **South-West Monsoon (summer monsoon) — June to September**
  - Winds blow **sea → land** across the Indian Ocean, the Arabian Sea and the Bay of Bengal
  - Cause: **unequal heating** — land heats faster → **low pressure** over the subcontinent, ocean relatively cooler → high pressure
  - Accounts for **most of India's annual rainfall**
- **North-East Monsoon (winter monsoon) — October to February**
  - Land cools faster → **high pressure** over land, low over seas
  - **Cold, dry winds blow land → sea** — generally no rain for most of India
  - **Exception:** picking up moisture over the Bay of Bengal → rain in the **eastern coast: Tamil Nadu, Andhra Pradesh, parts of Karnataka**

✍️ **EXAM POINTS — IMPORTANCE + ANCIENT KNOWLEDGE + MISSIONS**

- **Why monsoon matters:** most of India's **agriculture** depends on it (sowing, growing); good monsoon → food production + water in rivers, reservoirs, wells; affects daily life, transport, festivals, employment (especially rural).
- Excess rain → **floods**; weak monsoon → **droughts** — the economy, lifestyle and livelihoods swing with it.
- **Ancient rainfall science: Kṛiṣhiparāśhara** predicted rain from positions of the **Moon and Sun**; **Varāhamihira**'s Bṛihatsaṁhitā used **nakṣhatras** (lunar mansions — 27 divisions of the Moon's path; the Moon moves through one each day).
- **Kālidāsa**'s **Meghadūtam** (~5th century CE) mentions the date of monsoon onset over central India and traces the path of monsoon clouds.
- **National Monsoon Mission (NMM)** — Ministry of Earth Sciences; state-of-the-art weather and climate prediction models. **Mission Mausam** — aims to make India a global leader in weather/climate science, 'Weather Ready' and 'Climate Smart'.

T8 Climate Change + Carbon Footprint 🌱

☐ revised

📖 **DEFINITIONS**

Climate change — long-term changes in weather patterns (temperature, rainfall, wind) caused mainly by human activities — burning fossil fuels, deforestation, and industrial pollution. | **Carbon footprint** — the total amount of greenhouse gases released into the atmosphere as a result of human activities (energy use, transportation, production of goods and services).

💡 **SAMAJH (HINGLISH)**

Simple equation: **fuel jalaaya + ped kaate + factories chalayi** → greenhouse gases (CO_2, methane, nitrous oxide, water vapour) badhi → ye gases **heat trap** karti hain (car ki shisha-band gaadi jaise — dhoop me andar garmi!) → **global warming** → result: zyada floods, droughts, glaciers pighle, sea level badha, biodiversity khatre me. Solution bhi simple hai — **apna carbon footprint ghatao**: walk/cycle, lights band, paani bachao, plastic kam, ped lagao. Chhote steps, bada asar!

✍️ **EXAM POINTS**

- **Causes:** burning of fossil fuels, deforestation, industrial pollution releasing CO_2.
- **Greenhouse gases:** carbon dioxide, methane, nitrous oxide, water vapour — trap heat and raise global temperatures.
- **Effects:** more frequent floods and droughts, melting glaciers, rising sea levels, loss of biodiversity; threatens ecosystems, human health, agriculture, livelihoods — impacts **women and children** too.
- **Solutions:** reduce carbon footprints, use renewable energy, protect forests, adopt sustainable lifestyles — every small step counts.
- Activity (book): score your habits — transport (walk/cycle = low; flights often = very high), electricity, water use, waste & plastics — then take a **climate action pledge** with two changes.

T9 Case Study — Punjab Floods 2025 🌊

☐ revised

💡 **SAMAJH (HINGLISH)**

2025 me Punjab dooba kyun? **Double dhamaka:** natural (bahut heavy monsoon + **western disturbances** extra laaye; Himachal aur J&K me bhi baarish — Satluj, Beas, Ravi, Ghaggar pehle hi full thi!) + human-made (**dhūsī bāndh** — purane kamzor embankments toot gaye; log nadi ke bilkul paas bas gaye the; rivers/dams me **silt jama** tha — capacity kam; warnings late aayi). Result: fasal doobi, maringe, roads-bridges toote, bimariyan faili. Lesson: **planning hi bachai sakti hai**!

🌳 **SYSTEM — CAUSES (NATURAL + HUMAN-MADE) + EFFECTS**

- **Natural causes**
  - Very heavy monsoon rains intensified by **western disturbances** (extra moisture and rain)
  - Rain not only in Punjab but also in **Himachal Pradesh and Jammu & Kashmir**
  - Rivers **Satluj, Beas, Ravi (and Ghaggar)** already flowing high; extra water from hills + local rain → overflow
- **Human-made causes**
  - Weak, old river embankments (**dhūsī bāndh**) that could not hold the rising water
  - Houses and farms built **too close to rivers** → no space for floodwater to spread
  - **Silt and mud** collected in rivers and dams → reduced capacity
  - Late or unclear **flood warnings** → people unprepared
- **Effects**
  - Lives lost; thousands moved to **relief camps**
  - Farmland flooded; crops like **paddy** severely damaged; poultry and dairy farms destroyed; animals sick or dead
  - Roads, bridges, border fences and public buildings damaged
  - Murky standing water → **waterborne diseases** and sanitation problems; heavy economic losses and social disruption

✍️ **EXAM POINTS**

- Case shows the urgent need for improved **flood management and preparedness** (relief operations were carried out as water receded).
- Discussion points (book): natural factors vs human activities — which caused more? Could better planning have reduced damage? What are **NDMA** (National Disaster Management Authority) guidelines? What role can students/youth play?

T10 Quick Recap — Before We Move On ⚡

☐ revised

✍️ **EXAM POINTS — BOOK SUMMARY (SEEDHA SEEDHA)**

- The atmosphere is a mixture of gases — mainly **nitrogen and oxygen** — with small amounts of other gases, water vapour and dust that support life, form clouds and cause rainfall.
- Layers (bottom → top): **Troposphere, Stratosphere, Mesosphere, Thermosphere, Exosphere** — divided by changes in temperature and air density with height.
- **Weather** = daily atmospheric conditions; **climate** = average weather of a place over a long period (30+ years).
- India has a **tropical monsoon climate**; the IMD divides the year into **four seasons** — winter, summer, monsoon and post-monsoon.
- **Climate change** = long-term change in weather caused mainly by burning fuels and cutting trees → global warming and extreme weather; we reduce our **carbon footprint** by saving energy, using renewables, planting trees and eco-friendly living.

⚖️ Differences — Pakka Aane Wale

Basis	Weather	Climate
Meaning	Hour-to-hour, day-to-day conditions of the atmosphere	Sum total of weather conditions over a large area for an extended period

Time period	Short term (hours/days) — varies significantly day-to-day	Long term — usually **thirty years or more**
Example	"It will rain today"	"India has a tropical monsoon climate"

Basis	Troposphere	Stratosphere
Height	Average ~12 km (lowest layer)	Extends up to 50 km (above troposphere)
Temperature	**Decreases** with altitude	**Rises** with altitude
Key contents	Air we breathe, most water vapour and clouds	The **ozone layer** — filters harmful UV radiation
Activity	Nearly all weather phenomena (rain, fog, hail)	**Ideal for aeroplanes** — free of clouds and weather disturbances
Boundary	Separated from stratosphere by the **tropopause**	Separated from mesosphere by the **stratopause**

Basis	Land Breeze	Sea Breeze
Time	**Night**	**Day** (especially afternoon)
Direction	Land → sea	Sea → land
Reason	Land cools faster than the sea at night	Land becomes relatively warmer than the sea by day
Pressure	High pressure over land	Low pressure over land
Speed	Low (small temperature/pressure difference)	Relatively stronger

Basis	South-West Monsoon	North-East Monsoon
Season	Summer monsoon — **June to September**	Winter monsoon — **October to February**
Direction	Sea → land (Indian Ocean, Arabian Sea, Bay of Bengal)	Land → sea
Nature	Warm and **moist** — brings most of India's rainfall	Cold and **dry** for most of India
Pressure	Low pressure over land (unequal heating — land hotter)	High pressure over land (land cools faster)
Exceptional rain	Whole of India benefits	Picks moisture over Bay of Bengal → rain in **Tamil Nadu, Andhra Pradesh, Karnataka** (eastern coast)

📖 Definition Bank — Chapter 3 (likhne ke liye ready)

Atmosphere — blanket of air around the Earth, held by gravity; mixture of gases vital for life.

Altitude — height of a place above mean sea level (sea level = zero).

Weather — hour-to-hour and day-to-day conditions of the atmosphere.

Humidity — presence of water vapour in the air; warm air holds more of it.

Atmospheric (air) pressure — pressure exerted by the weight of air; highest at sea level.

Land breeze — local wind, land → sea at night (land cools faster).

Monsoon — seasonal reversal in wind direction during a year (Arabic: mausim = season).

Nakṣhatras — 27 equal divisions of the Moon's path across the sky; Moon crosses one each day.

Carbon footprint — total greenhouse gases released by human activities.

IMD — India Meteorological Department; recognises 4 seasons in India.

Dhūsī bāndh — weak/old river embankments that failed during the Punjab floods.

Gravity — fundamental force of attraction between objects with mass or energy.

Insolation — incoming solar energy from the Sun intercepted by the Earth.

Climate — average weather over a large area for 30+ years.

Precipitation — condensed water vapour falling due to gravity: drizzle, rain, snow, sleet, hail.

Wind — movement of air from high-pressure to low-pressure areas; named after the direction it blows from.

Sea breeze — local wind, sea → land in the day (land heats up more).

Seasons (ṛtu) — divisions of the year based on generalised monthly atmospheric conditions.

Climate change — long-term change in weather patterns, mainly due to human activities.

Greenhouse gases — CO_2, methane, nitrous oxide, water vapour; trap heat, raise temperatures.

Western disturbances — weather systems that brought extra moisture and rain in the Punjab 2025 floods.

NMM / Mission Mausam — MoES missions for monsoon prediction & 'Weather Ready, Climate Smart' India.

✅ MCQ Challenge — 10 Questions (Ch 3)

🎯 Solve karne ke baad score bharo → ____ / 10

Q1. The two most abundant gases in the atmosphere are —

a. oxygen and carbon dioxide
b. nitrogen and oxygen
c. nitrogen and argon
d. oxygen and ozone

Q2. Aeroplanes fly mainly in the stratosphere because it is —

a. the closest layer to Earth
b. free of clouds and weather disturbances
c. the hottest layer
d. full of ozone for fuel

Q3. Most meteorites burn up in the —

a. troposphere
b. stratosphere
c. mesosphere
d. exosphere

Q4. The colourful displays of light near the poles, called auroras, occur in the —

a. thermosphere
b. troposphere
c. mesosphere
d. stratosphere

Q5. Climate is the average weather of a place taken over a period of —

a. one year
b. ten years
c. thirty years or more
d. one month

Q6. Temperature decreases with increasing altitude ONLY in the —

a. troposphere and stratosphere
b. troposphere and mesosphere
c. stratosphere and thermosphere
d. mesosphere and exosphere

Q7. The wind blowing from the west is called a westerly. Winds are named after —

a. the direction they blow towards
b. the direction they blow from
c. their speed
d. the season of their birth

Q8. During the south-west monsoon, winds blow from sea to land because —

a. the landmass heats faster, creating low pressure over the subcontinent
b. the ocean heats faster than the land
c. gravity pulls the winds landward
d. the Himalayas block the winds

Q9. The north-east (winter) monsoon causes rainfall mainly in —

a. Punjab and Haryana
b. Tamil Nadu, Andhra Pradesh and parts of Karnataka
c. Rajasthan and Gujarat
d. the entire Gangetic plain

Q10. In the Punjab floods of 2025, the rivers involved were —

a. Ganga, Yamuna and Son
b. Satluj, Beas and Ravi
c. Godavari, Krishna and Kaveri
d. Narmada, Tapi and Mahanadi

🔑 Answer Key — Chapter 3

1 → b | 2 → b | 3 → c | 4 → a | 5 → c | 6 → b | 7 → b | 8 → a | 9 → b | 10 → b

- **1:** Nitrogen 78% + oxygen 21% = primary, most abundant gases.
- **2:** Stratosphere is free of clouds and other weather disturbances → ideal for flying.
- **3:** Most meteorites entering from space burn up in the mesosphere (up to 80 km).
- **4:** Auroras occur in the thermosphere (80–700 km); Borealis (north), Australis (south).
- **5:** Climate = average conditions over an extended period, usually 30+ years.
- **6:** Temperature falls with height only in the troposphere and the mesosphere.
- **7:** Winds are named after the direction they blow FROM (westerly = from the west).
- **8:** Summer = land heats faster than sea → low pressure over land → moist sea winds move in.
- **9:** NE winds pick moisture over the Bay of Bengal → eastern coast (TN, AP, Karnataka) gets rain.
- **10:** Punjab rivers: Satluj, Beas, Ravi (plus Ghaggar) swelled after heavy rains.

Big Questions — Exam-Ready Answers (Ch 3)

Q1. What is the atmosphere? Explain its composition.

ANSWER

- The atmosphere is a **blanket of air surrounding the Earth**, pulled down around it by **gravity** — a mixture of gases vital for the survival of all living beings.
- It **shields us from the Sun's harmful radiation** (including ultraviolet) and **regulates the Earth's temperature** by trapping some of the Sun's energy.
- Composition: **nitrogen 78%**, **oxygen 21%**, argon 0.93%, carbon dioxide 0.04%, and others 0.03% (helium, neon, krypton, xenon, ozone, hydrogen).
- It also contains **water vapour (0.1–0.4%)** — important for cloud formation and precipitation — and tiny dust particles.
- The composition **varies with altitude**.

Q2. Describe the structure (layers) of the atmosphere.

ANSWER

- The atmosphere has a layered structure defined by changes in **temperature and density** with altitude (density is highest near the surface).
- **Troposphere** (avg ~12 km): temperature decreases with height; the air we breathe, most water vapour and clouds; nearly all weather phenomena; bounded above by the tropopause.
- **Stratosphere** (up to 50 km): ideal for aeroplanes — free of clouds and weather; contains the **ozone layer** that filters UV radiation; bounded by the stratopause.
- **Mesosphere** (up to 80 km): temperature decreases with height; most **meteorites burn up** here.
- **Thermosphere** (80–700 km): temperature rises rapidly (absorbs X-rays and short-wave UV); helps **radio transmission**; the ionosphere is part of it; **auroras** occur here.
- **Exosphere**: uppermost, very thin air; light gases like helium and hydrogen escape into space.

Q3. In which layer of the atmosphere do aeroplanes fly and why?

ANSWER

- Aeroplanes fly in the **stratosphere**, the second layer extending up to **50 km** above the troposphere.
- It is **free of clouds and other weather disturbances** (rain, storms, turbulence), making flights smooth and safe.
- A notable feature is the **layer of ozone gas**, which shields us by filtering the Sun's harmful ultraviolet radiation.
- Weather phenomena are confined to the troposphere below, so the stratosphere offers a stable corridor for long-distance flying.

Q4. Distinguish between (a) the troposphere and stratosphere, (b) the south-west and north-east monsoon.

ANSWER

- **(a) Height:** troposphere ~12 km vs stratosphere up to 50 km.
- **(a) Temperature:** falls with altitude in the troposphere but rises in the stratosphere.
- **(a) Contents:** troposphere has the air we breathe, most water vapour, clouds and **all weather**; stratosphere has the **ozone layer** and is ideal for aeroplanes.
- **(b) Season:** SW monsoon = June–September (summer); NE monsoon = October–February (winter).
- **(b) Direction and nature:** SW blows **sea → land**, warm and moist — brings most of India's rain; NE blows **land → sea**, cold and dry — but brings rain to the **eastern coast (Tamil Nadu, Andhra Pradesh, Karnataka)** after crossing the Bay of Bengal.

Q5. Explain the elements of weather and climate.

ANSWER

- **Temperature:** varies day–night and season-wise; controlled by **insolation**, which decreases from the equator to the poles — so temperature also decreases poleward.
- **Humidity:** water vapour in the air; warmer air holds more vapour; humid days slow down drying of clothes and evaporation of sweat.
- **Precipitation:** saturated air condenses and falls as drizzle, rain, snow, sleet or hail; factors: prevailing winds, mountains, seasons; rain is the most common form.
- **Atmospheric pressure:** weight of air; highest at sea level, falls with altitude; warm air rises → low pressure (cloudy/wet); cold air sinks → high pressure (clear/sunny).
- **Wind:** air moving from high-pressure to low-pressure areas — from a gentle breeze to storm force.

Q6. Why do we not feel the pressure of the atmosphere? What is the relation between pressure and wind?

ANSWER

- The air above us exerts significant force, but **air presses on us from all sides** and our bodies exert an **equal counter-pressure** — the two cancel out, so we feel nothing.
- Pressure is **highest at sea level** and decreases with altitude; horizontally it depends on the air's temperature.

- In high-temperature areas, heated air **rises** → creates **low pressure** — associated with cloudy skies and wet weather.
- In colder areas, air is heavy and **sinks** → **high pressure** — associated with clear, sunny skies.
- **Wind** is simply air moving from high-pressure to low-pressure areas — its direction is influenced by the Earth's rotation.

Q7. Describe the four main seasons of India recognised by the IMD.

ANSWER

- **Winter (December to early April):** coldest months are December–January; average temperature ~10–15°C in the north-west, rising to ~20–25°C towards the south-east.
- **Summer or pre-monsoon (April to June):** hottest month is April in western and southern regions, May in the north; averages 32–40°C across inland India.
- **Monsoon or rainy season (June to September):** dominated by the humid **south-west summer monsoon**, which sweeps across the country from late May or early June; it recedes from north India from early October, with South India receiving more rainfall then.
- **Post-monsoon or retreating monsoon (October to December):** October–November are usually cloudless in north-west India.
- The climate of India is broadly **tropical monsoon**; Himalayan states also experience autumn and spring.

Q8. What is monsoon? Explain the mechanism of the south-west monsoon.

ANSWER

- Monsoon is the **seasonal reversal in wind direction** during a year; Arab traders named it from the Arabic word **mausim** ('season'), having noticed the reversal on sailing voyages.
- **Step 1:** During summer, the Indian landmass **heats up faster** than the surrounding oceans.
- **Step 2:** This creates a **low-pressure area** over the Indian subcontinent, while the Indian Ocean stays cooler with **high pressure**.
- **Step 3:** Winds move from high to low pressure — so **moist winds blow from the ocean towards the land** across the Indian Ocean, the Arabian Sea and the Bay of Bengal (June–September).
- **Step 4:** These moisture-laden winds bring **rainfall** — the SW monsoon accounts for most of the rainfall in the country throughout the year.

Q9. What is climate change? What are its causes and effects?

ANSWER

- Climate change refers to **long-term changes in weather patterns** — temperature, rainfall and wind.
- **Causes:** mainly human activities — burning of fossil fuels, deforestation, and industrial pollution releasing gases like carbon dioxide.
- These increase **greenhouse gases** (CO_2, methane, nitrous oxide, water vapour) that trap heat and raise global temperatures.
- **Effects:** more frequent floods and droughts, melting glaciers, rising sea levels, loss of biodiversity; impacts human health, agriculture and livelihoods — including women and children.
- **Solutions:** reduce carbon footprints, use renewable energy, protect forests, adopt sustainable lifestyles — every small step counts.

Q10. Explain the causes and effects of the Punjab floods of 2025.

ANSWER

- **Natural causes:** very heavy monsoon rains intensified by **western disturbances**; rain also fell in Himachal Pradesh and Jammu & Kashmir; the rivers **Satluj, Beas and Ravi** were already flowing high before the downpour began.
- **Human-made causes:** weak and old embankments (**dhūsī bāndh**); houses and farms built too close to rivers; silt and mud reducing the capacity of rivers and dams; late or unclear flood warnings.
- **Effects:** loss of lives; thousands shifted to relief camps; farmland and crops like paddy damaged; poultry and dairy farms destroyed; animals sick or dead.
- Roads, bridges, border fences and public buildings were damaged; standing murky water spread **waterborne diseases**.
- The floods underscored the urgent need for improved **flood management and preparedness** — better planning could have reduced the damage.

You're exam-ready! 💪🧠

Definitions ratto → system trees revise karo (iske under kya tha!) → differences ke tables yaad karo → MCQs solve karke score check karo → big Q&A se answer-writing practice. Geography = sorted. All the best! 🍀